

Pneumonia Detection from Chest X-Rays

Overview

This project builds a binary image classifier to detect pneumonia from chest X-rays. The core objective was to develop a model that could reliably flag pneumonia cases while keeping false negatives as low as possible. In a clinical screening context, missing a positive diagnosis carries a much higher cost than a false alarm, so recall on the pneumonia class was treated as the primary metric throughout the project.

Three model backbones were evaluated: ResNet34, ResNet50, and DenseNet121. All were trained using fastai with ImageNet pretrained weights, fp16 mixed precision, and one-cycle learning rate scheduling. The final model achieved 97% accuracy, 99% pneumonia recall, and a 98% F1 score on a held-out test set that was not touched until all modeling decisions had been finalized.

Data

The dataset was sourced from the mnenendezg/pneumonia_x_ray dataset on HuggingFace, which is derived from the Kaggle chest X-ray pneumonia dataset. It contains labeled chest X-ray images across two classes: normal and pneumonia. The dataset came pre-split into train, validation, and test partitions, but those splits contained overlapping images across partitions, which would have introduced data leakage into the evaluation.

To address this, all three splits were concatenated into a single dataframe and deduplicated by hashing the raw image bytes using Polars. This ensured that identical images were removed regardless of which split they came from or what filename they were stored under. Images that shared hash values but had different file paths were confirmed to have consistent labels before being dropped. After deduplication, the dataset was reshuffled with a fixed random seed and resplit 70/20/10 across train, validation, and test sets. Images were renamed to their hash values for consistent identification and saved to disk as PNGs.

The final cleaned dataset had a roughly 3:1 class imbalance in favor of PNEUMONIA. This is consistent with real-world clinical distributions where patients presenting for chest X-rays tend to skew toward positive diagnoses, so no resampling was applied to the class distribution.

Modeling

Setup

All models were trained using fastai's vision_learner with ImageNet pretrained weights and fp16 mixed precision. One-cycle learning rate scheduling was used throughout, with learning rates identified using fastai's lr_find and manually validated against the loss curve rather than relying solely on the automated suggestions, which were found to occasionally misfire.

Data augmentation was applied using fastai's aug_transforms with parameters loosely based on those reported in published pneumonia detection literature. Horizontal and vertical flips were disabled since chest X-rays have a fixed anatomical orientation, and other transforms included up to 20 degrees of rotation, zoom up to 1.2x, warp up to 0.2, and lighting variation up to 0.2. All

images were resized to 224x224, and a fixed random seed was set both globally and in the DataLoaders to ensure reproducibility across runs.

SaveModelCallback was used during all training runs to capture the best validation loss checkpoint rather than defaulting to the final epoch. This was important because validation loss frequently ticked back up in later epochs even as training loss continued to fall, meaning the last epoch was rarely the best one.

Model Comparison

Three different pretrained models were used in this project. Each was initially trained with pretrained weights frozen, training only the classification head. Where applicable, the backbone was then unfrozen for additional fine-tuning cycles with a discriminative learning rate slice. The learning rates were selected from lr_find curves for both the frozen and unfrozen phases.

Model	Val Loss	Accuracy	Recall	F1
ResNet34	0.189	93.5%	92.3%	95.4%
ResNet50	0.169	94.4%	94.0%	96.1%
DenseNet121 (frozen only)	0.163	96.3%	95.8%	97.4%

As seen in the above results, DenseNet121 outperformed both ResNet variants across every metric. The most notable finding during model comparison was that unfreezing the DenseNet121 backbone consistently degraded performance. Multiple unfrozen training runs were attempted with different learning rate configurations, and none of them improved on the frozen best checkpoint. This suggests that the pretrained features transferred well to chest X-ray classification without further modification.

The initial DenseNet121 training runs were also problematic. Early attempts showed high validation loss in the first epoch due to a misplaced lr_find suggestion that placed the steep point at an extremely small learning rate around 1e-6, well below the optimal range visible in the loss curve. After manually selecting a learning rate from the curve and retraining, the model converged more cleanly and ultimately outperformed both ResNet variants.

ResNet50 had the lowest validation loss among the ResNets but produced worse pneumonia recall than ResNet34, suggesting it was more conservative in its predictions and more likely to miss positive cases. For this task, that tradeoff was not acceptable, making ResNet34 the stronger of the two ResNet variants despite its higher overall loss.

Inference Configuration

For the final iteration of the model, TTA (test-time augmentation) was used to improve the performance of the model. TTA generates multiple augmented versions of each input image, runs each through the model independently, and then averages the resulting predictions. This tends to produce more robust probability estimates.

Threshold tuning was also explored by testing classification thresholds from 0.2 to 0.5 and comparing precision, recall, and F1 at each value. The results were largely flat across

thresholds, which indicates the model produces highly confident predictions rather than clustering around the decision boundary. Based on these results, the default 0.5 threshold with TTA was selected as the final inference configuration.

Results

Final evaluation was performed once on the held-out test set using the frozen DenseNet121 checkpoint with TTA. No modeling decisions were made after this point. The test set had not been examined at any stage of the model selection or hyperparameter tuning process.

Class	Precision	Recall	F1	Support
NORMAL	0.97	0.93	0.95	147
PNEUMONIA	0.97	0.99	0.98	392
Overall (weighted)	0.97	0.97	0.97	539

Test performance was consistent with validation performance and in some metrics slightly better, which is a positive sign that the validation-based tuning process did not overfit to the validation set. The headline number is PNEUMONIA recall of 99%, meaning only 1% of positive cases were missed on data the model had never seen. NORMAL precision of 97% also indicates that very few healthy patients were incorrectly flagged.

Misclassification analysis on the validation set was performed by reviewing the highest-loss predictions visually, however, to my untrained eye, no systematic patterns were identified in the errors. The misclassified images did not share obvious visual characteristics that could be addressed through augmentation adjustments or preprocessing changes. This suggests the remaining failures might be ambiguity in the images themselves rather than a correctable issue with the modeling pipeline.

Future Work

A few directions worth exploring given more time and resources:

- Pretrained medical imaging backbones: models pretrained on large chest X-ray datasets like CheXpert or NIH ChestX-ray14 would bring domain-specific features into the starting weights rather than relying on ImageNet transfer. This could meaningfully improve performance, particularly on edge cases.
- Ensemble methods: combining predictions from DenseNet121 and ResNet50 could outperform either model alone. The two models make different types of errors, and averaging their probability outputs tends to reduce overall error rates.
- Incorporating imagery from other datasets and expanding the problem to a multiclass classification problem would greatly increase the value of this model, making it a useful aid in diagnosis of a wider variety of conditions that can be observed through chest radiographs. Additionally, radiographs from other sources would add variety to the images, increasing the robustness and generalizability of the model.

Data Source

Dataset: mnenendezg/pneumonia_x_ray via HuggingFace Datasets